TerraMind on SmallMinesDS: Final Report

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Summary

We compared a frozen, pretrained TerraMind v1 tiny encoder with the same encoder trained from scratch for mine segmentation on SmallMinesDS 2022, using 1 to 1,491 (all) training images, in two experiments, and set both against a plain UNet of the same size in a third. **Experiment 1** trained every run for a fixed 50 epochs under four conditions (each encoder with and without D4 augmentation, 196 runs): augmentation improved the frozen encoder in every seed (+0.02 to +0.04 IoU) but had no consistent effect on the trainable one, so only the augmented conditions were kept. **Experiment 2** replaced the fixed budget with early stopping on test IoU and recorded the cost of every run (98 runs). In both experiments the frozen pretrained encoder gives the higher IoU at every N ≥ 10 in every seed (all images: 0.745 vs 0.673 in Experiment 1, 0.747 vs 0.688 in Experiment 2), and with 100 images it exceeds what the trainable encoder reaches with all 1,491. Early stopping gives the same IoU as the fixed 50 epochs from N = 250 on and 0.01–0.03 higher IoU at N = 3–100. The frozen encoder is also cheaper to train: 2.4 vs 5.5 GFLOP per training sample, and with all images 0.172 vs 0.370 GPU hours until early stopping. Experiment 3 trained a plain UNet with the same number of parameters from scratch under the Experiment 2 protocol: it beat both TerraMind models at every N and in every seed (all images: 0.789; one image: 0.579 vs 0.402), at 6.8 times the FLOPs per training sample of the frozen encoder.

Setup (all experiments)

Item	Setting
Data	SmallMinesDS 2022, 128 × 128 px patches; official split: 1,491 training, 644 test
Inputs	Sentinel-2 L2A (10 bands) + Sentinel-1 RTC (VV, VH, in dB); DEM not used
Model	Experiments 1–2: TerraMind v1 tiny encoder (5.9 M parameters) + shallow UNet decoder (1.0 M), built with TerraTorch. Experiment 3: plain UNet (6.8 M)

Item	Setting
Encoder conditions	Pretrained, frozen (only the decoder learns) / random init, trainable; Experiment 3: random init, trainable
Augmentation	D4: random rotation by 0/90/180/270°, with or without mirroring
Training images N	1, 3, 5, 10, 20, 50, 100, 250 (5 seeds); 500, 1000, all = 1,491 (3 seeds)
Optimiser	AdamW, weight decay 0.01, batch size min(8, N)
Metric	Mine-class IoU on the 644 test patches; mean ± std over seeds
Hardware	NVIDIA GeForce RTX 3080 Laptop GPU (8 GB)

Within a seed, the conditions use the same training images, decoder initialisation and batch order, so they are directly comparable. Code, results and reproduction steps: github.com/asimarslan11/SmallMinesDS-Experiment-Project.

Experiment 1: fixed training budget, four conditions

Every run trained for exactly 50 epochs with a cosine learning rate from 5e-3 to 1e-5, and the final model was evaluated; 4 conditions × 49 (seed, N) pairs = 196 runs, 5.0 GPU hours in total. The frozen pretrained encoder is ahead at every N ≥ 10 in every seed, and augmentation adds +0.02 to +0.04 IoU to it at every N ≥ 3 (5/5 seeds), while for the trainable encoder the effect changes sign with N. With 10 images, the frozen encoder with augmentation (0.591) matches the trainable encoder with 250 images (0.584–0.605).

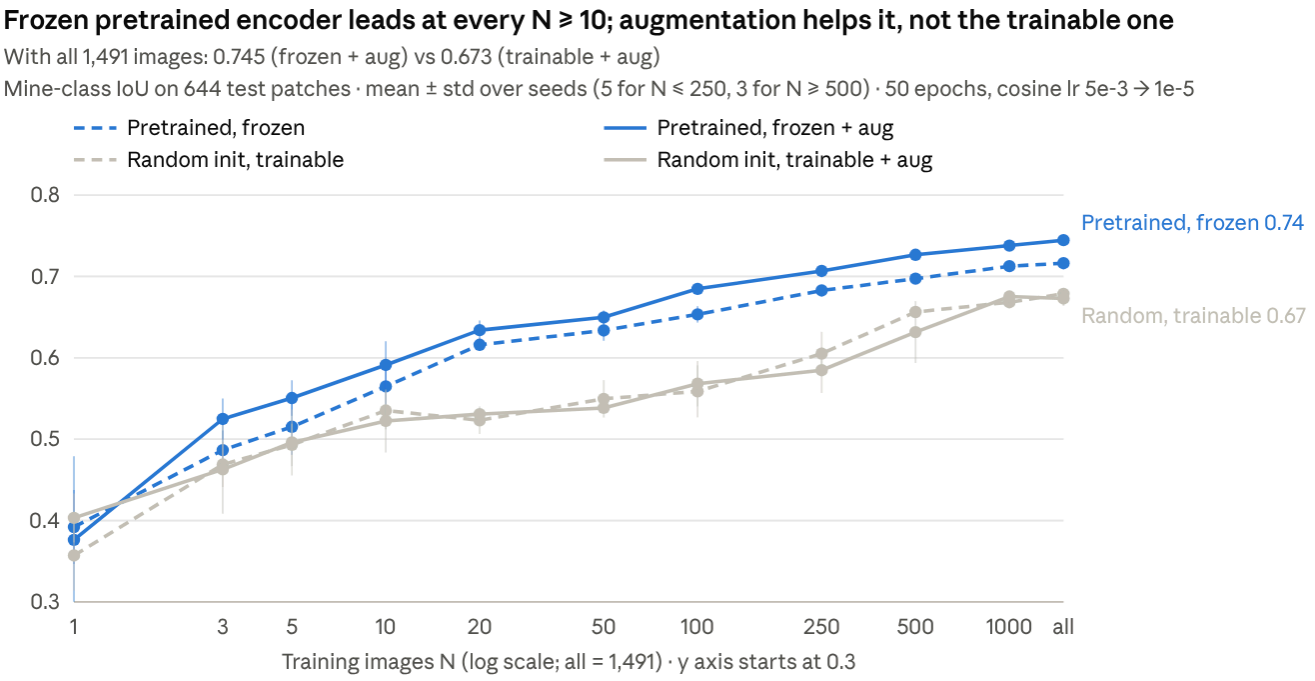


Figure 1. Experiment 1: mine-class IoU on the test set against the number of training images, mean ± std over seeds

Table 1. Experiment 1 — mine-class IoU on the test set (mean ± std over seeds).

N	seeds	Pretrained, frozen	Pretrained, frozen + aug	Random init, trainable	Random init, trainable + aug
1	5	0.392 ± 0.046	0.376 ± 0.103	0.357 ± 0.043	0.403 ± 0.030
3	5	0.487 ± 0.024	0.525 ± 0.025	0.469 ± 0.028	0.463 ± 0.054
5	5	0.515 ± 0.034	0.551 ± 0.022	0.493 ± 0.026	0.496 ± 0.040
10	5	0.565 ± 0.023	0.591 ± 0.029	0.535 ± 0.023	0.522 ± 0.039
20	5	0.616 ± 0.007	0.634 ± 0.012	0.523 ± 0.017	0.531 ± 0.007
50	5	0.633 ± 0.013	0.650 ± 0.008	0.550 ± 0.023	0.538 ± 0.005
100	5	0.653 ± 0.010	0.685 ± 0.007	0.559 ± 0.032	0.568 ± 0.028
250	5	0.683 ± 0.004	0.707 ± 0.002	0.605 ± 0.027	0.584 ± 0.028
500	3	0.697 ± 0.003	0.726 ± 0.003	0.656 ± 0.003	0.632 ± 0.038
1000	3	0.712 ± 0.003	0.738 ± 0.002	0.668 ± 0.006	0.675 ± 0.002
all (1,491)	3	0.716 ± 0.003	0.745 ± 0.003	0.679 ± 0.004	0.673 ± 0.009

Table 2. Experiment 1 — IoU gain from D4 augmentation (augmented minus unaugmented run with the same seed and training images; mean ± std, in brackets the seeds that improved).

N	Pretrained, frozen	Random init, trainable
1	−0.016 ± 0.073 (3/5)	+0.046 ± 0.044 (4/5)
3	+0.038 ± 0.016 (5/5)	−0.006 ± 0.045 (2/5)
5	+0.035 ± 0.027 (5/5)	+0.003 ± 0.034 (3/5)
10	+0.027 ± 0.010 (5/5)	−0.013 ± 0.024 (2/5)
20	+0.018 ± 0.007 (5/5)	+0.008 ± 0.017 (3/5)
50	+0.016 ± 0.005 (5/5)	−0.011 ± 0.024 (2/5)
100	+0.031 ± 0.007 (5/5)	+0.010 ± 0.007 (5/5)
250	+0.024 ± 0.003 (5/5)	−0.021 ± 0.034 (2/5)
500	+0.029 ± 0.002 (3/3)	−0.025 ± 0.036 (1/3)
1000	+0.025 ± 0.002 (3/3)	+0.007 ± 0.008 (2/3)
all (1,491)	+0.029 ± 0.005 (3/3)	−0.006 ± 0.014 (2/3)

Experiment 2: early stopping and training cost

Only the two augmented conditions were run (98 runs, 4.1 GPU hours in total). Instead of a fixed budget, each run trained until its test IoU stopped improving: test IoU was checked every 5 epochs for every N; training stopped once 20 epochs (4 checks) passed without a gain above 10^{-3} over the best check; the learning rate started at 5e-3 and was halved at every non-improving check (minimum 1e-5); the model at the best check is reported. Steps, FLOPs and GPU hours were recorded for every run.

The frozen pretrained encoder is ahead at every $N \geq 5$ in every seed, 0.747 vs 0.688 with all images, and with 100 images (0.694) it already exceeds the trainable encoder with all 1,491. IoU rises with N for both encoders and flattens for the frozen one from N = 500 (0.729, 0.740, 0.747). Compared with Experiment 1 (Table 1, augmented columns), the frozen encoder reaches 0.01–0.03 higher IoU at N = 3–100 and the same IoU (within 0.005) from N = 250 on; the trainable encoder is equal or up to 0.03 higher. The frozen encoder costs 2.4 GFLOP per training sample against 5.5 for the trainable encoder, and with all images it stops after 0.172 GPU hours (284 TFLOP) against 0.370 (837 TFLOP).

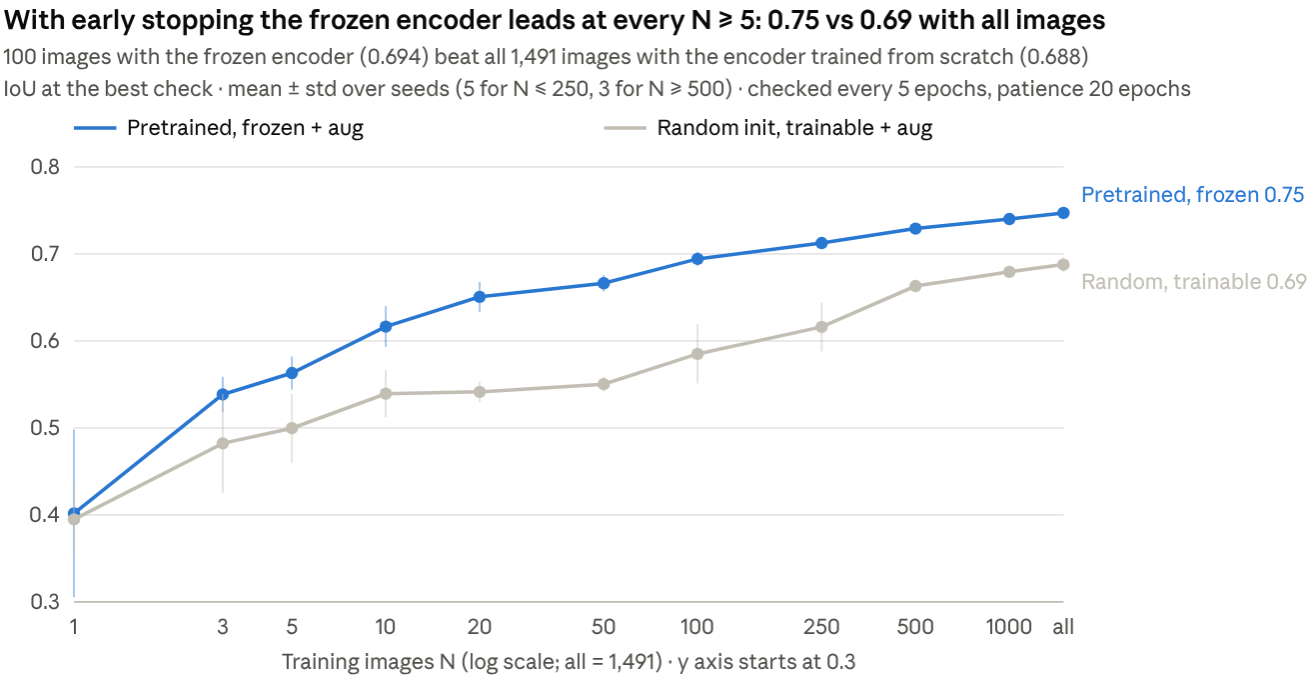


Figure 2. Experiment 2: mine-class IoU (best check) against the number of training images, mean ± std over seeds

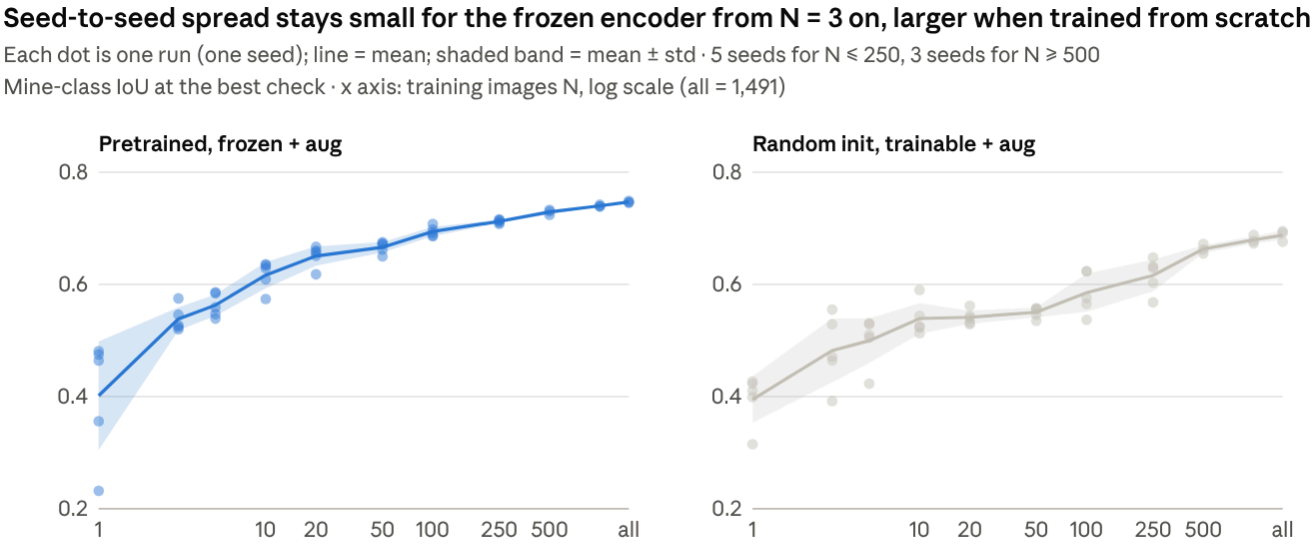


Figure 3. Experiment 2: seed-to-seed variance of IoU — every run as a dot, mean as a line, mean ± std as a shaded band

Table 3. Experiment 2 — mine-class IoU on the test set (best check; mean ± std over seeds).

N	seeds	Pretrained, frozen + aug	Random init, trainable + aug
1	5	0.402 ± 0.096	0.395 ± 0.041
3	5	0.539 ± 0.020	0.482 ± 0.057

N	seeds	Pretrained, frozen + aug	Random init, trainable + aug
5	5	0.563 ± 0.019	0.500 ± 0.040
10	5	0.616 ± 0.023	0.539 ± 0.027
20	5	0.651 ± 0.017	0.541 ± 0.012
50	5	0.666 ± 0.009	0.550 ± 0.009
100	5	0.694 ± 0.008	0.585 ± 0.034
250	5	0.712 ± 0.003	0.616 ± 0.028
500	3	0.729 ± 0.004	0.663 ± 0.007
1000	3	0.740 ± 0.002	0.680 ± 0.006
all (1,491)	3	0.747 ± 0.001	0.688 ± 0.008

Table 4. Experiment 2 — training cost until early stopping, pretrained, frozen + aug (mean ± std over seeds).

N	best epoch	stop epoch	steps	GPU hours	total TFLOP
1	43.0 ± 28.0	63.0 ± 28.0	63 ± 28	0.003 ± 0.001	15.4 ± 6.8
3	39.0 ± 9.2	59.0 ± 9.2	59 ± 9	0.002 ± 0.000	14.7 ± 2.3
5	55.0 ± 8.9	75.0 ± 8.9	75 ± 9	0.003 ± 0.000	19.0 ± 2.3
10	63.0 ± 10.8	83.0 ± 10.8	166 ± 22	0.005 ± 0.001	22.0 ± 2.9
20	76.0 ± 13.9	96.0 ± 13.9	288 ± 42	0.006 ± 0.001	27.7 ± 4.0
50	65.0 ± 17.0	85.0 ± 17.0	595 ± 119	0.010 ± 0.002	30.6 ± 6.1
100	57.0 ± 6.8	77.0 ± 6.8	1001 ± 88	0.015 ± 0.001	36.9 ± 3.2
250	62.0 ± 14.4	82.0 ± 14.4	2624 ± 459	0.034 ± 0.006	68.4 ± 12.0
500	53.3 ± 15.5	73.3 ± 15.5	4620 ± 974	0.059 ± 0.012	104.7 ± 22.1
1000	53.3 ± 4.7	73.3 ± 4.7	9167 ± 589	0.116 ± 0.007	191.8 ± 12.3
all (1,491)	55.0 ± 7.1	75.0 ± 7.1	14025 ± 1322	0.172 ± 0.017	283.5 ± 26.7

Table 5. Experiment 2 — training cost until early stopping, random init, trainable + aug (mean ± std over seeds).

N	best epoch	stop epoch	steps	GPU hours	total TFLOP
1	34.0 ± 23.5	54.0 ± 23.5	54 ± 24	0.002 ± 0.001	13.3 ± 5.8
3	49.0 ± 28.5	69.0 ± 28.5	69 ± 29	0.003 ± 0.001	17.8 ± 7.4

N	best epoch	stop epoch	steps	GPU hours	total TFLOP
5	40.0 ± 11.4	60.0 ± 11.4	60 ± 11	0.003 ± 0.001	16.1 ± 3.1
10	25.0 ± 17.6	45.0 ± 17.6	90 ± 35	0.004 ± 0.002	13.4 ± 5.2
20	35.0 ± 27.9	55.0 ± 27.9	165 ± 84	0.005 ± 0.002	19.4 ± 9.8
50	39.0 ± 5.8	59.0 ± 5.8	413 ± 41	0.010 ± 0.001	30.6 ± 3.0
100	49.0 ± 26.3	69.0 ± 26.3	897 ± 342	0.021 ± 0.008	54.9 ± 21.0
250	52.0 ± 19.1	72.0 ± 19.1	2304 ± 612	0.047 ± 0.012	117.2 ± 31.1
500	73.3 ± 12.5	93.3 ± 12.5	5880 ± 786	0.122 ± 0.015	281.3 ± 37.6
1000	75.0 ± 25.5	95.0 ± 25.5	11875 ± 3187	0.247 ± 0.069	549.7 ± 147.5
all (1,491)	78.3 ± 8.5	98.3 ± 8.5	18388 ± 1589	0.370 ± 0.031	836.6 ± 72.3

GPU hours are the wall-clock time of each run (model build, training and every test check). FLOPs are counted with torch.utils.flop_counter per sample (2.37 GFLOP per training sample for the frozen encoder, which has no backward pass through the encoder; 5.54 GFLOP for the trainable one; 1.87 GFLOP per evaluated sample) and multiplied by the samples processed; the total includes every test check (644 images each).

Experiment 3: plain UNet of the same size

The third model is a standard UNet with no TerraMind component, trained from scratch: 6.83 M parameters (encoder 4.15 M, decoder 2.68 M), within 1 % of the 6.90 M of the TerraMind models. Its encoder halves the 128 × 128 input four times down to an 8 × 8 bottleneck, the same grid as TerraMind's tokens (channels 30/60/120/240/480, two 3 × 3 convolutions with BatchNorm and ReLU per stage); the decoder upsamples with transposed convolutions and skip connections. Data, inputs, training images, seeds, augmentation and the early-stopping protocol are exactly those of Experiment 2 (49 runs, 1.9 GPU hours).

The UNet gives the highest IoU at every N and in every seed, ahead of both TerraMind models: 0.789 vs 0.747 (frozen) and 0.688 (trainable) with all images, and 0.579 vs 0.402 with a single image. With 10 images (0.687) it is level with the frozen TerraMind encoder at 100 images (0.694), and with 250 images (0.753) it exceeds the frozen encoder with all 1,491. The gain costs compute: 16.1 GFLOP per training sample against 2.4 for the frozen encoder, so with all images it uses 1,806 vs 284 TFLOP and 0.227 vs 0.172 GPU hours.

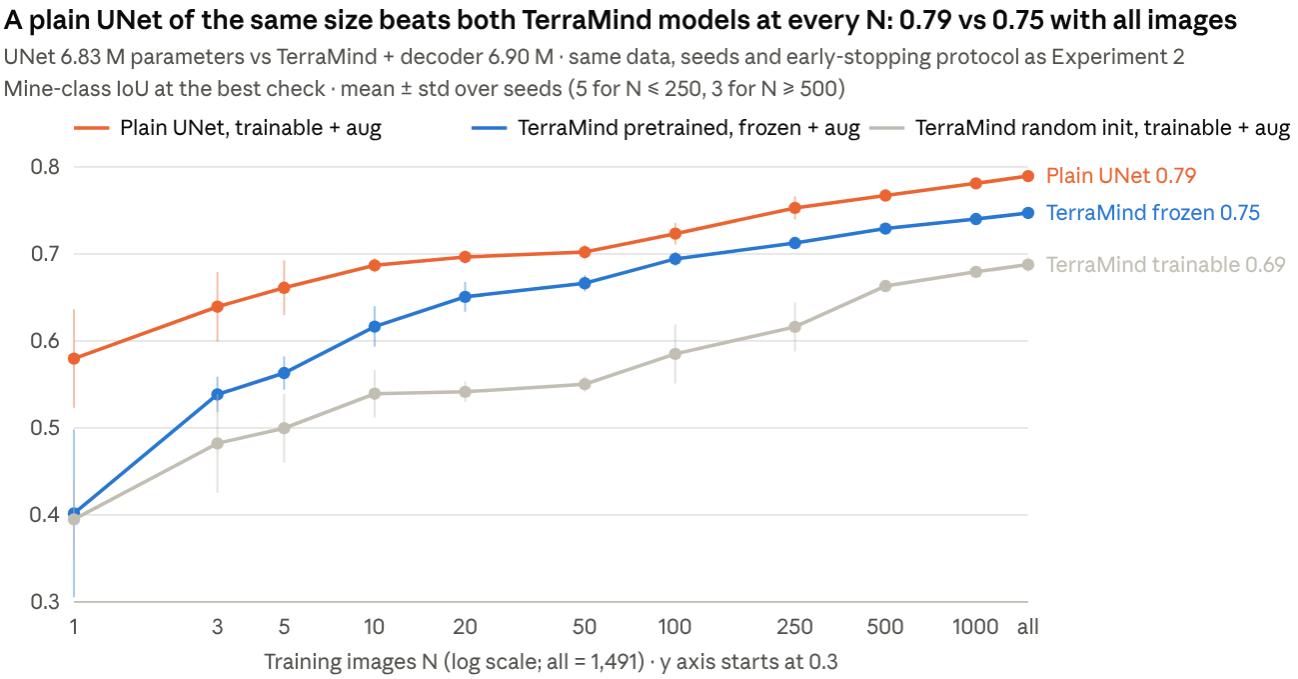


Figure 4. Experiment 3: mine-class IoU (best check) of the plain UNet against the two TerraMind models of Experiment 2, mean ± std over seeds

Table 6. Experiment 3 — mine-class IoU on the test set (best check; mean ± std over seeds; the TerraMind columns repeat Table 3).

N	seeds	Plain UNet, trainable + aug	TerraMind pretrained, frozen + aug	TerraMind random init, trainable + aug
1	5	0.579 ± 0.057	0.402 ± 0.096	0.395 ± 0.041
3	5	0.639 ± 0.040	0.539 ± 0.020	0.482 ± 0.057
5	5	0.661 ± 0.031	0.563 ± 0.019	0.500 ± 0.040
10	5	0.687 ± 0.008	0.616 ± 0.023	0.539 ± 0.027
20	5	0.696 ± 0.008	0.651 ± 0.017	0.541 ± 0.012
50	5	0.702 ± 0.009	0.666 ± 0.009	0.550 ± 0.009
100	5	0.723 ± 0.012	0.694 ± 0.008	0.585 ± 0.034
250	5	0.753 ± 0.013	0.712 ± 0.003	0.616 ± 0.028
500	3	0.767 ± 0.005	0.729 ± 0.004	0.663 ± 0.007
1000	3	0.781 ± 0.001	0.740 ± 0.002	0.680 ± 0.006
all (1,491)	3	0.789 ± 0.003	0.747 ± 0.001	0.688 ± 0.008

Table 7. Experiment 3 — training cost until early stopping, plain UNet + aug (mean ± std over seeds; 16.06 GFLOP per training sample, 5.39 GFLOP per evaluated sample).

N	best epoch	stop epoch	steps	GPU hours	total TFLOP
1	54.0 ± 7.3	74.0 ± 7.3	74 ± 7	0.003 ± 0.000	52.5 ± 5.2
3	56.0 ± 15.9	76.0 ± 15.9	76 ± 16	0.004 ± 0.001	56.4 ± 11.8
5	59.0 ± 30.7	79.0 ± 30.7	79 ± 31	0.004 ± 0.002	61.2 ± 23.8
10	46.0 ± 11.6	66.0 ± 11.6	132 ± 23	0.005 ± 0.001	56.4 ± 9.9
20	60.0 ± 15.2	80.0 ± 15.2	240 ± 45	0.007 ± 0.001	81.2 ± 15.4
50	59.0 ± 9.2	79.0 ± 9.2	553 ± 64	0.011 ± 0.001	118.2 ± 13.7
100	68.0 ± 18.9	88.0 ± 18.9	1144 ± 245	0.021 ± 0.005	202.3 ± 43.4
250	63.0 ± 12.9	83.0 ± 12.9	2656 ± 412	0.045 ± 0.007	390.7 ± 60.7
500	55.0 ± 7.1	75.0 ± 7.1	4725 ± 445	0.079 ± 0.008	654.1 ± 61.7
1000	55.0 ± 4.1	75.0 ± 4.1	9375 ± 510	0.150 ± 0.007	1256.2 ± 68.4
all (1,491)	53.3 ± 10.3	73.3 ± 10.3	13713 ± 1921	0.227 ± 0.042	1806.3 ± 253.1